

Part1

Independent Variable: Race and Sex

dependent Variable: Highest year of school completed

H0₁: The mean highest year of school completed is equal across racial groups

H0₂: The mean highest year of school completed is equal across gender groups

H0₃: There is no interaction effect between race and gender on highest year of school completed.

H1₁: The mean highest year of school completed is differs across racial groups

H1₂: The mean highest year of school completed is differs across gender groups

H1₃: There is a significant interaction effect between race and gender on highest year of school completed.

Levene's Test of Equality of Error Variances^{a,b}

		Levene Statistic	df1	df2	Sig.
Highest year of school completed	Based on Mean	8.811	5	2339	<.001
	Based on Median	8.724	5	2339	<.001
	Based on Median and with adjusted df	8.724	5	1913.716	<.001
	Based on trimmed mean	8.853	5	2339	<.001

Tests the null hypothesis that the error variance of the dependent variable is equal across groups.

a. Dependent variable: Highest year of school completed

b. Design: Intercept + RACE + SEX + RACE * SEX

Tests of Between-Subjects Effects

Dependent Variable: Highest year of school completed

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	458.884 ^a	5	91.777	10.586	<.001
Intercept	228296.512	1	228296.512	26334.025	<.001
RACE	441.450	2	220.725	25.461	<.001
SEX	.901	1	.901	.104	.747
RACE * SEX	6.581	2	3.291	.380	.684
Error	20277.399	2339	8.669		
Total	462913.000	2345			
Corrected Total	20736.283	2344			

a. R Squared = .022 (Adjusted R Squared = .020)

Result:.

Levene's Test of Homogeneity of Variances was significant, ($p < 0.001$), indicating that the assumption of homogeneity of variance was violated, A two-way ANOVA was conducted to examine the effects of race and sex on highest year of school completed. The analysis showed a significant main effect of race, $F = 25.461$, $p < 0.001$, indicating that educational attainment differed significantly across racial groups. However, there was no significant main effect of Sex ($P = 0.747$) and Race*Sex ($P = 0.684$). Overall, race was associated with significant differences in educational attainment, whereas sex and the interaction between race and sex were not

statistically significant.

Test Assumptions:

A two-way ANOVA was appropriate for this analysis because only three independent groups were compared (White vs. Black vs. Other) on a continuous dependent variable (Highest year of school completed). The dependent variable was assumed to be approximately normally distributed within each group. Finally, Levene's Test was significant ($p < 0.001$), indicating that the assumption of homogeneity of variance was violated,

Multiple Comparisons

Dependent Variable: Highest year of school completed

Bonferroni

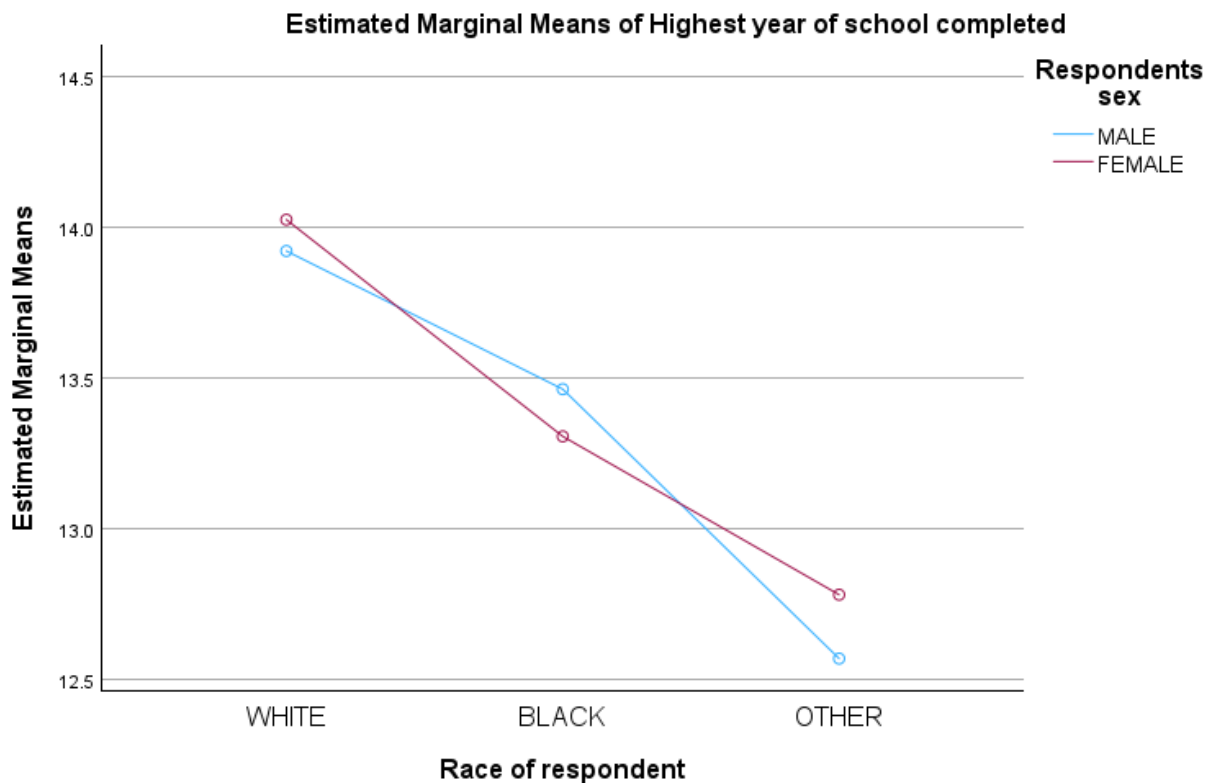
(I) Race of respondent	(J) Race of respondent	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
WHITE	BLACK	.61 [*]	.166	<.001	.21	1.01
	OTHER	1.30 [*]	.193	<.001	.83	1.76
BLACK	WHITE	-.61 [*]	.166	<.001	-1.01	-.21
	OTHER	.69 [*]	.234	.010	.13	1.25
OTHER	WHITE	-1.30 [*]	.193	<.001	-1.76	-.83
	BLACK	-.69 [*]	.234	.010	-1.25	-.13

Based on observed means.

The error term is Mean Square(Error) = 8.669.

*. The mean difference is significant at the .05 level.

Post hoc Bonferroni comparisons showed significant differences among all racial groups in highest year of school completed. White respondents reported significantly more years of schooling than Black respondents (mean difference = 0.61, $p < 0.001$) and Other respondents (mean difference = 1.30, $p < 0.001$). Black respondents also reported significantly more years of schooling than Other respondents (mean difference = 0.69, $p = 0.010$). These findings indicate a clear pattern in educational attainment: White respondents had the highest average years of schooling, followed by Black respondents, and then Other respondents.



The interaction plot further showed that educational attainment generally decreased from White to Black to Other racial groups for both males and females. Although the male and female lines slightly crossed in the interaction plot, the differences were small, and the interaction effect between race and sex was not statistically significant, $F = 0.380$, $p = 0.684$, and small gender differences were visible within some racial groups, these differences were minimal and not statistically significant ($p = 0.747$).

Part 2

The purpose of this multi-way ANOVA exercise was to examine whether race and sex, as well as their interaction, had significant effects on the highest year of school completed. Multi-way ANOVA compares the mean differences of a continuous dependent variable across multiple categorical independent variables and evaluates both main effects and interaction effects using the F-ratio, which is calculated by dividing the variance between groups by the variance within groups. In this analysis, Levene's Test was significant ($p < 0.001$), indicating that the assumption of homogeneity of variance was violated, so the results should be interpreted with caution. The results showed a significant main effect of race, $F(2, 2339) = 25.461$, $p < 0.001$, indicating that educational attainment differed significantly across racial groups. Post hoc comparisons revealed that White respondents reported the highest average years of schooling, followed by Black respondents, and then Other respondents. However, there was no significant main effect of sex, $F(1, 2339) = 0.104$, $p = 0.747$, and no significant interaction effect between race and sex, $F(2, 2339) = 0.380$, $p = 0.684$, suggesting that the relationship between race and educational attainment was consistent across males and females.